

UNIVERSITY OF DHAKA

DataLex: Explainable SIEM Security Intelligence Using Generative AI

Submitted by

Arpita Paul

Exam Roll: 303

Registration No: H-7

Abu Syeed Sajid Ahmed

Exam Roll: 325

Registration No: H-26

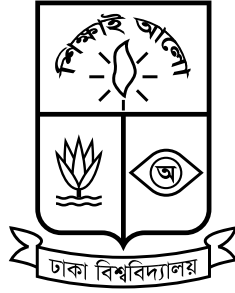
Submitted to the

Department of Computer Science and Engineering

University of Dhaka, Bangladesh

in partial fulfillment of the requirements for the degree of
Professional Masters in Information and Cyber Security (PMICS).

Declaration Form



University of Dhaka

We hereby declare that the work in this report titled “**DataLex: Explainable SIEM Security Intelligence Using Generative AI**” has been carried out by us under the supervision of Abu Ahmed Ferdaus and co-supervision of Mehedi Hasan. We further declare that where information has been derived from the literature, we have adequately cited and referenced the original sources. No part of this report was previously presented for another degree or diploma at this or any other institution.

Name of the Candidate

Arpita Paul

Name of the Candidate

Abu Syeed Sajid Ahmed

Abu Ahmed Ferdaus

Associate Professor

Dept. of

Computer Science and Engg.

University of Dhaka

Major (retd) Md Mehedi Hasan

Co-instructor

Dept. of

Computer Science and Engg.

University of Dhaka

“Common sense is like deodorant. The people who need it most never use it.”

Bill Murray

UNIVERSITY OF DHAKA

Abstract

DataLex represents an innovative Security Information and Event Management (SIEM) solution leveraging offline generative AI to address contemporary cybersecurity challenges. By integrating an advanced language model (LLAMA 3.0) with sophisticated correlation algorithms, the system provides comprehensive threat detection and analysis across diverse technological ecosystems, including national infrastructure, financial technology, healthcare, and urban systems.

The research introduces a groundbreaking privacy-preserving framework operating entirely offline, eliminating external data transmission risks. Key innovations include zero external API reliance, localized event processing, and transparent AI decision-making. The methodology encompasses developing an offline generative AI architecture, creating an interpretable security analysis approach, and rigorously validating system performance.

DataLex's primary contributions include the first offline, generative AI-powered SIEM system, a novel explainable AI threat detection methodology, and a privacy-centric threat intelligence architecture. By addressing systemic limitations of existing technologies – such as excessive false positives and limited contextual understanding – the project advances cybersecurity research and offers a robust solution for organizations seeking comprehensive, privacy-focused threat monitoring.

Acknowledgements

Thesis research emerges from intricate academic collaborations, and our DataLex project epitomizes this intellectual synergy. The profound academic mentorship of Abu Ahmed Ferdaus (Associate Professor of Computer Science and Engineering, University of Dhaka) and Major (retd) Md Mehedi Hasan transcended conventional supervisory boundaries, transforming our initial cyber-security and AI conceptualization into a rigorous, innovative research endeavor. Their complementary expertise – one anchored in advanced computational methodologies, the other in complex security architectures – provided the critical intellectual infrastructure that enabled us to navigate the multifaceted challenges of developing an offline generative AI-powered SIEM system, consistently challenging our theoretical assumptions and refining our research methodology.

Contents

Declaration Form	i
Abstract	iii
Acknowledgements	iv
List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Project Domain and Social Significance	1
Emerging Application Areas:	2
1.2 Work Domain Definition and Systemic Interactions	3
1.2.1 System Components and Architecture	3
1.2.2 Privacy and Security Considerations	4
1.2.3 Societal Impact	4
1.3 Problem Statement	4
1.4 Motivation and Research Needs	5
1.5 Research Objectives	6
1.6 Conceptual Framework	6
1.6.1 Experimental Design	7
1.6.2 Technical Implementation	8
1.6.3 Evaluation Frameworks	8
1.7 Key Contributions	10
1.8 Thesis Organization	10
2 Related Works	11
2.1 Related Concepts and Chapter Knowledge	11
2.2 Comparative Discussions on Existing Works	14
2.3 Experimental Observations and Recent Developments	15
2.3.1 Recent Advances in ML for SIEM	15
2.3.2 Impact of Open-Source SIEM	15
2.3.3 Role of Containerization	15
2.3.4 Threat Intelligence Integration	16
2.4 Limitations and Research Gaps in Existing SIEM Solutions	16

3	Proposed Methodologies	17
3.1	Overview	17
3.2	System Model and Assumptions	18
3.2.1	System Components and Their Interactions	19
3.2.2	Key Assumptions	19
3.3	Proposed Method	22
3.3.1	SIEM Components and their purposes	22
3.3.1.1	SIEM	23
3.3.1.2	Wazuh – Open-Source SIEM and Security Monitoring	23
3.3.1.3	Suricata – Intrusion Detection and Prevention System	24
3.3.1.4	Wazuh Dashboard – Data Visualization and Dash- board	24
3.3.1.5	AI-Driven Threat Intelligence (GPT Service)	25
3.3.1.6	Retrieval-Augmented Generation (RAG) – AI-Powered Search	25
3.3.1.7	FAISS – Facebook AI Similarity Search	26
3.3.1.8	The LLAMA (Large Language Model Meta AI) . . .	26
3.3.1.9	DeepSeek Qwen	27
3.3.1.10	Choosing Between LLAMA and DeepSeek Qwen . .	27
3.3.1.11	Integration and Workflow	28
3.3.2	Any Secondary Contribution	29
3.4	Conclusion	29
4	Performance Evaluation	31
4.1	Simulation or Testbed Environment	31
4.1.1	System Configuration	32
4.1.2	Test Scenarios and Methodology	32
4.2	Performance Metrics	33
4.2.1	Performance Evaluation Methodology	34
4.2.2	Performance Results	35
4.2.2.1	Inference Time	35
4.2.2.2	Retrieval Latency	35
4.2.2.3	Memory Usage	36
4.2.2.4	Throughput	36
4.2.2.5	Token Generation Rate	37
4.2.3	Wazuh and Elasticsearch Performance Metrics	38
4.2.4	System-Wide Resource Utilization Metrics	39
4.2.5	System Resource Monitoring Commands	39
4.2.6	Security and Compliance Metrics	41
4.2.6.1	Security Evaluation Commands	41
4.3	Results Analysis	42
4.3.1	Performance Comparisons and Trend	42
4.3.1.1	The Inference Time Analysis	42
4.3.1.2	Insights on Retrieval Latency	43

4.3.1.3	Memory Usage Patterns	44
4.3.1.4	Throughput Performance Analysis	45
4.3.1.5	Token Generation Rate Evaluation	45
4.3.2	Comparative Analysis of Intrusion Detection Systems (Snort vs. Suricata)	46
4.3.3	Comparative Analysis of SIEM Implementations	46
4.4	Conclusion	47
5	Conclusions	49
5.1	Summary of Work	50
5.2	Future Direction	50
5.2.1	Expanding Threat Intelligence Sources	51
5.2.2	Dynamic AI Model Updates and Fine-Tuning	51
5.2.3	Integration of Anomaly Detection and Predictive Analysis	51
5.2.4	Multi-Language Support and Accessibility	52
5.2.5	AI-Guided Automated Incident Response	52
5.2.6	Improving Explainability and Trust in AI-Generated Insights	52
5.2.7	Enhancing the Chat-Based Security Interaction Model	52
5.2.8	Cloud and Hybrid Deployment Models	53
5.2.9	Evolving into an AI-Powered Security Governance Platform	53
5.2.10	AI-Driven Security Compliance and Risk Management	53
5.2.11	Conclusion	54
	Bibliography	55

List of Figures

3.1	High level diagram of the proposed model	22
4.1	Inference Time Across Hardware Configurations	35
4.2	Retrieval Latency Across Hardware Configurations	36
4.3	Memory Usage Across Hardware Configurations	37
4.4	Throughput Across Hardware Configurations	38
4.5	Token Generation Rate Across Hardware Configurations	38

List of Tables

2.1	Comparison of Existing SIEM Systems and Proposed Solution . . .	14
4.1	Performance Metrics and Outcomes	39
4.2	System Performance Metrics and Outcomes	40
4.3	Security and Compliance Metrics	41
4.4	Inference Time Across Hardware Configurations	43
4.5	Retrieval Latency for Different Hardware	43
4.6	Memory Usage for Different Hardware	44
4.7	Throughput for Different Hardware	45
4.8	Token Generation Rate for Different Hardware	45
4.9	Comparison of Snort and Suricata Features	47
4.10	Comparative Analysis of SIEM Solutions	48

Chapter 1

Introduction

In the age of digital transformation, cyber-security has emerged as a critical frontier, presenting complex challenges that require groundbreaking solutions to safeguard vital technological infrastructure and sensitive data. This chapter delves into the core principles, driving motivations, and strategic vision of the DataLex project, comprehensively addressing the urgent demand for sophisticated threat detection and comprehensive management systems in an increasingly vulnerable digital landscape.

1.1 Project Domain and Social Significance

Traditional security measures are no longer sufficient due to the increase in sophisticated cyber threats brought about by the growth of digital infrastructure. Modern cybersecurity relies heavily on Security Information and Event Management (SIEM), which makes it possible to monitor and respond to security problems in real time. A sophisticated Generative AI-driven SIEM solution like DataLex is now necessary for proactive threat detection and intelligent threat analysis due to the growing complexity of cloud computing, IoT, and distributed networks.

This project improves cybersecurity resilience by protecting important digital assets and sensitive data. DataLex offers contextualized threat insights through the

integration of Generative AI into SIEM, facilitating automated, human-like threat assessments that increase the effectiveness of incident response. This method makes security intelligence more widely available while lowering dependency on specialized security experts. Increasing cyber defenses helps businesses avoid financial losses, promotes trust in online transactions, and guarantees adherence to international cybersecurity laws.

Emerging Application Areas: The potential applications of enhanced SIEM solutions go far beyond typical enterprise cybersecurity. Emerging domains requiring sophisticated threat identification and analysis include:

- **Critical Infrastructure Protection:** Protecting national power grids, transportation systems, healthcare networks, and telecommunications infrastructure from potential cyber-physical attacks.
- **Financial Technology (FinTech) Security:** Securing complicated financial transaction systems, blockchain networks, and digital payment platforms against sophisticated financial crime and cyber espionage.
- **Healthcare Cyber-security:** Ensuring the integrity and confidentiality of sensitive medical records, safeguarding medical devices, and adhering to strict data protection standards.
- **Smart City Ecosystems:** Monitoring and protecting integrated urban infrastructure, such as intelligent transportation systems, municipal services, and public utility networks.
- **Advanced Research and Academic Institutions:** Defending sensitive research data, intellectual property, and collaborative research environments against targeted cyber-attacks.

1.2 Work Domain Definition and Systemic Interactions

Effective cyber security demands a nuanced, interdisciplinary approach that examines the intricate web of system interactions, technological architectures, and complex digital relationships. This section offers an in-depth examination of the DataLex ecosystem, unveiling its sophisticated technical framework and elucidating the advanced mechanisms that power intelligent, contextual threat detection and response.

1.2.1 System Components and Architecture

DataLex is a sophisticated SIEM ecosystem made up of numerous technical components that work together to provide complete, intelligent threat detection and analysis. The system architecture includes the following:

- **Data Collection Interfaces:** Stable systems for gathering log data from a variety of sources, such as network devices, servers, applications, and security endpoints of the organization.
- **LLAMA 3.0 Processing Engine:** An advanced, offline large language model capable of processing complex security events, producing human-readable threat analysis, and giving contextual information.
- **Threat Correlation and Anomaly Detection Module:** Advanced algorithms that discover potential security incidents by examining complex links between seemingly unrelated events.
- **Interactive Dashboard:** A user-friendly interface that allows security analysts to explore, examine, and respond to recognized threats via natural language interaction.

1.2.2 Privacy and Security Considerations

The DataLex system is being rigorously designed to address significant privacy concerns inherent in modern SIEM solutions. By working totally offline and using an open-source language model, the solution avoids the risks involved with communicating sensitive security data to external servers.

Key privacy measures include but are not limited to:

- Complete data sovereignty
- No external API dependency
- Localized processing of security events
- Transparent and explainable AI decision-making
- Compliance with strict data protection requirements

1.2.3 Societal Impact

The creation of DataLex goes beyond technical innovation, making a substantial contribution to societal cybersecurity resilience. By democratizing advanced threat detection capabilities and lowering reliance on proprietary, cloud-based solutions, the technology enables enterprises of all sizes to employ sophisticated security monitoring.

1.3 Problem Statement

Despite significant technological advancements, contemporary Security Information and Event Management (SIEM) systems remain fundamentally limited. This section critically examines the systemic challenges that current security technologies struggle to overcome, setting the stage for innovative solutions.

- **High False Positive Rates:** Existing rule-based systems generate numerous, frequently unrelated notifications, causing alert fatigue among security experts.
- **Limited Contextual Understanding:** Traditional SIEM solutions struggle to understand the subtle, contextual relationships between seemingly unrelated security incidents.
- **Dependency on External Services:** The majority of advanced threat detection systems rely on cloud-based APIs, which pose substantial privacy and latency issues.
- **Lack of Interoperability:** Most AI-powered security technologies function as “black boxes,” with little transparency into their reasoning and decision-making processes.
- **Scalability Issues:** Existing solutions frequently fail to react dynamically to changing threat landscapes and organizational infrastructure changes.

1.4 Motivation and Research Needs

The impetus for DataLex derives from a critical analysis of current SIEM limitations and the increasing promise of generative AI technology. Traditional security monitoring systems have reached a technical inflection point.

Cyber threats are becoming more complex, attack tactics are becoming more sophisticated, computational needs for complete threat analysis are increasing, and security mechanisms need to be more intelligent and adaptive.

DataLex tackles several systemic issues by incorporating an offline, open-source large language model into the SIEM framework:

- Providing transparent, understandable threat intelligence
- Removing privacy risks related to external API dependencies

- Enabling natural language interaction with security event data
- Developing a flexible, adaptive threat detection mechanism

1.5 Research Objectives

Strategic research requires clear, defined goals that drive innovation and address critical technological gaps. This section outlines the precise objectives of the DataLex project, providing a comprehensive roadmap for developing an advanced, privacy-preserving security intelligence system. The key research aims of the DataLex project are:

- **Architectural Innovation:** Creating a full SIEM system that utilizes offline generative AI capabilities.
- **Privacy Enhancement:** Showcasing a threat detection mechanism that ensures entire data sovereignty.
- **Explainability Development:** Developing an AI-powered security analysis approach with strong interpretability.
- **Performance Evaluation:** Thoroughly evaluating the proposed system's efficiency, accuracy, and comparative advantage.
- **Scalability Validation:** Demonstration of the system's adaptability to different organizational contexts.

1.6 Conceptual Framework

Rigorous scientific inquiry demands a structured, multidisciplinary approach. This section details the comprehensive methodology employed in the DataLex project,

explaining the experimental design, technical implementation strategies, and evaluation frameworks that guided the research. The research process for DataLex takes a broad, multidisciplinary approach.

The proposed framework depicts the combination of SIEM and GPT services to provide an advanced, offline, and privacy-preserving threat detection solution. The SIEM service includes components like Logstash, Wazuh, Suricata, and the Kibana Dashboard to collect, process, and visualize security data. The GPT service leverages LLAMA 3.0 and other modules to analyze logs, generate responses, and enable user interaction through an LLM Chat UI. The overall system integrates traditional SIEM tools with a novel GPT-based approach, aiming to deliver a comprehensive, privacy-focused, and explainable security monitoring solution.

1.6.1 Experimental Design

This sub-section provides a detailed overview of the experimental design for the proposed system, focusing on the integration of modern SIEM technologies with advanced generative AI techniques.

- **SIEM Service:** This microservice incorporates core SIEM components, including Wazuh, Logstash, Elasticsearch, and Kibana, along with Suricata, an Intrusion Detection System (IDS). It handles the essential operations of data collection, processing, and visualization.
- **GPT Service:** Developed using the Retrieval-Augmented Generation (RAG) framework, this microservice employs Python, Flask, LangChain, Ollama, and a vector database (FAISS). It utilizes embedding models and LLAMA 3.0 as the generative AI model to provide intelligent threat analysis and contextual insights. It also provides APIs to manage a knowledge base, enabling the generative AI to improve response accuracy by leveraging stored knowledge.

- **Scheduler Service:** A Python-based microservice designed to periodically fetch logs from Suricata, vectorize them using an embedding model, and store them in the vector database (FAISS).

1.6.2 Technical Implementation

The technical implementation of the DataLex system revolves around addressing key challenges in cybersecurity using advanced technologies and tailored solutions such as:

- **Tuning LLAMA 3.0 for Cybersecurity:** The LLAMA 3.0 generative AI model has been fine-tuned to address specific cybersecurity scenarios, ensuring contextual relevance and precision in threat analysis.
- **Custom Threat Correlation Algorithms:** Novel algorithms have been designed to enhance the system's ability to correlate diverse security events, improving the accuracy of threat detection.
- **Interactive Dashboard Design:** An intuitive and user-friendly dashboard interface has been developed to provide real-time insights, actionable recommendations, and enhanced usability for security teams.

1.6.3 Evaluation Frameworks

The evaluation framework outlines the methodologies and metrics employed to assess the efficiency, scalability, and reliability of the proposed system. It encompasses key aspects such as retrieval performance, dataset preparation, comparative analysis, scalability metrics, explainability, and continuous improvement, ensuring a comprehensive assessment of the system's capabilities and practical applicability.

- **Retrieval Component:** Metrics such as query latency and response latency serve as indicators of efficiency, with query latency evaluating the time required to retrieve documents and response latency capturing the duration from input query to output response generation.
- **Dataset Preparation:** The preparation of datasets is critical, ensuring alignment between retrieval and generation datasets with real-world scenarios. Evaluation datasets are sourced or constructed with human-validated queries, documents, and expected outputs to maintain relevance and accuracy.
- **Comparative Analysis:** A systematic comparison with other open-source models highlights relative performance and identifies advantages and limitations of the proposed approach.
- **Scalability and Deployment Metrics:** Scalability is assessed by analyzing latency under varying loads, alongside resource utilization metrics, including CPU/GPU usage and memory consumption. Offline compatibility is evaluated to validate the system's functionality in environments lacking internet connectivity.
- **Explainability and Interpretability:** Explainability is facilitated through attribution analysis tools like SHAP or LIME, enabling validation of document relevance and generation choices. Retrieval traceability ensures that retrieved documents are logged for verification, while attention analysis visualizes how retrieval context influences the generative process.
- **Feedback and Continuous Learning:** The system incorporates feedback loops to refine retrieval accuracy and response quality iteratively. Continuous monitoring of model drift ensures the timely integration of newer data, enhancing the system's adaptability over time.

1.7 Key Contributions

The DataLex project introduces groundbreaking innovations to the field of cybersecurity, contributing to the research domain in several significant ways:

- **Offline Generative AI-Powered SIEM:** Development of the first-ever SIEM system powered by generative AI that operates entirely offline, ensuring data sovereignty and privacy.
- **Explainable Threat Detection:** A novel approach to enhance threat detection by leveraging explainable AI, making security insights more accessible and actionable for security professionals.
- **Large Language Models for Security Analysis:** Application of large language models (e.g., LLAMA 3.0) to perform advanced security incident analysis and threat intelligence.
- **Privacy-Preserving Architecture:** Creation of an architecture that ensures privacy while incorporating generative AI and threat intelligence, addressing modern data protection challenges.

1.8 Thesis Organization

The following outline provides a structural overview of the project: Chapter 2 (Related Works) explores existing research on SIEM solutions, intrusion detection systems, and AI-driven security analytics, identifying gaps that DataLex addresses. Chapter 3 (Proposed Methodologies) details the system architecture, integration strategies, and the application of Generative AI for threat detection. Chapter 4 (Performance Evaluation) presents benchmarking results, analyzing system efficiency in log retrieval, inference time, and security event processing. Chapter 5 (Conclusion) summarizes the research contributions, discusses key findings, and suggests potential improvements for Generative-AI integrated SIEM solutions in the future.

Chapter 2

Related Works

Numerous research initiatives in intrusion detection, cybersecurity monitoring, and AI-driven threat intelligence have impacted the development of AI-enhanced SIEM systems. Traditional SIEM solutions, IDS/IPS frameworks, and the incorporation of machine learning for anomaly detection have all been the subject of several research efforts. This chapter examines the main current approaches, their drawbacks, and how DataLex expands on them to provide a security monitoring system that is more effective, scalable, and enhanced by AI.

2.1 Related Concepts and Chapter Knowledge

The integration of Generative AI in cybersecurity systems, particularly in Intrusion Detection Systems (IDS), has been explored as a means to overcome the limitations of traditional SIEM and IDS platforms, which often rely on signature-based detection and predefined rules. These traditional methods typically result in high false-positive rates and limited adaptability to emerging threats, making them less effective in detecting novel or sophisticated attack patterns [1]. ChatIDS addresses these issues by utilizing Generative AI to provide human-readable explanations for IDS alerts, improving the interpretability of alerts for security analysts and reducing the cognitive load associated with manual analysis. This advancement aligns with

the growing need for more explainable and user-friendly security tools in modern organizations, as traditional systems often overwhelm analysts with large volumes of technical data and false positives [1].

The integration of SIEM systems with data lakes and AI/ML enhances traditional SIEM platforms, which often struggle with scalability and detecting complex threats. Traditional SIEM relies on signature-based detection, leading to high false positives and limited adaptability. By combining SIEM with data lakes, organizations can manage both structured and unstructured data more effectively, improving scalability and flexibility. The use of AI and machine learning algorithms boosts real-time detection, automates responses, and enables advanced analytics, significantly improving threat identification and response accuracy [2].

The growing sophistication of cyber threats necessitates the evolution of SIEM systems beyond traditional rule-based approaches [3]. AI-driven threat intelligence enhances SIEM capabilities by leveraging machine learning and real-time analytics to detect, classify, and respond to cyber threats with greater accuracy. This AI-powered integration improves SIEM's ability to identify zero-day vulnerabilities, reduce false positives, and automate incident response mechanisms. In industries like finance, healthcare, and government, AI-enhanced SIEM has shown practical benefits. However, challenges such as computational power requirements and the balance between automation and human intervention still exist. Future trends, including self-learning AI models and AI-assisted cyber forensics, are expected to further enhance real-time cybersecurity monitoring and response.

AI-driven threat intelligence is revolutionizing the field of cybersecurity by improving the capabilities of real-time threat detection, analysis, and response. Traditional security systems often rely on static, rule-based approaches that struggle to keep up with the increasingly sophisticated nature of cyber threats. The integration of machine learning models – including supervised and unsupervised learning, reinforcement learning, and natural language processing (NLP) – enhances the effectiveness of threat detection systems by allowing them to dynamically adapt to new attack patterns [4]. AI frameworks are particularly valuable for addressing the growing

complexity of cyber threats, enabling systems to not only detect but also predict and respond to potential security incidents. As AI tools continue to evolve, they aim to provide a proactive, dynamic approach to cybersecurity that can continuously improve its ability to combat emerging threats, overcoming many of the limitations of traditional methods.

The integration of Generative AI and Large Language Models (LLMs) into cybersecurity is enhancing real-time threat detection and response [5]. Traditional security systems struggle to keep up with sophisticated threats, but LLMs like GPT-4, BERT, and LLaMA provide advanced capabilities in malware detection, intrusion prevention, and threat intelligence analysis. Despite their promise, LLMs introduce vulnerabilities such as prompt injection and adversarial instructions. Mitigation strategies, including Reinforcement Learning with Human Feedback (RLHF) and Retrieval-Augmented Generation (RAG), along with techniques like Half-Quadratic Quantization (HQQ), help optimize model performance and address challenges in training LLMs with cybersecurity datasets, aiming to improve defense mechanisms against evolving threats.

As network-based attacks continue to rise, Intrusion Detection Systems (IDS) remain crucial for detecting malicious traffic. While many Machine Learning-based IDS models exist, they often struggle with accurately detecting and classifying rare or uncommon attack types. A hybrid approach that uses synthetic data generated by Generative Adversarial Networks (GANs) to train a Deep Reinforcement Learning (DRL) model, trained on the NSL-KDD dataset for IDS evaluation, demonstrates that synthetic data generated by GANs improves the performance of the DRL model, particularly in correctly classifying minority classes, compared to training on imbalanced real-world datasets [6].

The increasing sophistication of cybersecurity threats demands more advanced solutions. Traditional SIEM systems struggle with large data volumes and complex threats, leading to the exploration of next-generation systems that integrate data lakes and AI. Data lakes provide scalable storage for structured and unstructured

data, while AI enables real-time anomaly detection, predictive analytics, and proactive threat mitigation. AI-driven SIEM systems improve threat detection speed, accuracy, and efficiency, reducing false positives and operational costs. However, challenges like data integration complexity, high costs, and privacy concerns persist. Emerging technologies such as edge AI and quantum computing are expected to further shape the future of cybersecurity [7].

2.2 Comparative Discussions on Existing Works

Table 2.1 provides a comparative analysis of existing Security Information and Event Management (SIEM) systems, highlighting the distinctions between traditional, modern, and the proposed solution. This comparison evaluates critical features such as cost, scalability, threat detection, ease of use, integration, and incident response capabilities, showcasing the advancements offered by the proposed approach.

Feature	Traditional SIEMs	Modern SIEMs	Proposed Solution
Cost	High (license fees, maintenance)	Reduced for open-source solutions	Cost-effective with open-source tools (Wazuh)
Scalability	Limited for large-scale environments	Improved with containerization	Dynamic scaling via Docker
Threat Detection	Rule-based, prone to false positives	ML-enhanced, better anomaly detection	Hybrid: Rules + ML for reduced false positives
Ease of Use	Requires expertise for deployment	User-friendly dashboards (e.g., Kibana)	Simplified interface + AI-powered contextual help
Integration	Limited to specific platforms	Better cloud and application integrations	Customized for organizations
Incident Response	Manual or semi-automated	Automation frameworks integrated	AI-driven automation and workflow suggestions

TABLE 2.1: Comparison of Existing SIEM Systems and Proposed Solution

2.3 Experimental Observations and Recent Developments

The field of cybersecurity is characterized by rapid technological evolution and emerging research paradigms. This section explores recent experimental observations and technological developments that are reshaping our understanding of security information management, highlighting key advancements and their potential implications.

2.3.1 Recent Advances in ML for SIEM

Studies show that incorporating supervised learning reduces false positives by 30 percent compared to traditional rule-based SIEM systems. However, training time and resource consumption are major drawbacks.

2.3.2 Impact of Open-Source SIEM

Tools like Wazuh and Suricata provide significant cost savings for small to medium organizations, with comparable performance to proprietary systems. Limited documentation and community support are the primary concerns.

2.3.3 Role of Containerization

Deploying SIEM systems in Docker containers improves scalability and portability, making it ideal for dynamic workloads. However, security risks inherent to containers must be addressed.

2.3.4 Threat Intelligence Integration

Proactive threat detection requires sophisticated intelligence integration strategies. This subsection examines the emerging approaches to incorporating dynamic threat intelligence feeds into security management systems, exploring both the opportunities and challenges of this innovative approach.

2.4 Limitations and Research Gaps in Existing SIEM Solutions

Current SIEM systems face challenges such as high costs in traditional models and limited enterprise support in open-source options. While modern SIEMs improve scalability and integration through containerization, they still struggle with efficient machine learning implementation, requiring significant resources. Additionally, manual or semi-automated incident response processes and suboptimal cloud and application integration remain key gaps. These limitations highlight the need for cost-effective, scalable, AI-driven solutions tailored to specific organizational requirements, like banking environments.

Chapter 3

Proposed Methodologies

This chapter outlines the architectural framework, implementation approach, and key innovations of DataLex. By integrating SIEM technologies with Generative AI, the system enhances security monitoring, enabling advanced threat detection and interactive intelligence for security teams.

3.1 Overview

In the evolving landscape of cybersecurity, organizations face a continuous threat from sophisticated attacks, necessitating the development of advanced threat detection and mitigation mechanisms. Traditional Security Information and Event Management (SIEM) systems provide real-time monitoring, detection, and incident response capabilities, but often lack explainability and contextual awareness when dealing with large-scale security events. Meanwhile, recent advancements in Generative AI (GenAI) have introduced powerful models that can enhance threat analysis through context-aware intelligence.

This research introduces DataLex, a novel SIEM-GenAI integrated system designed to enhance cybersecurity monitoring and response by combining traditional SIEM functionalities with Generative AI capabilities. The system employs Elasticsearch, Wazuh-dashboard, Filebeat, and Suricata to collect, process, and visualize security

data, while LLAMA 3.0, FAISS, and LangChain enable AI-driven threat analysis and user interaction via a conversational assistant.

The primary objective of this work is to bridge the gap between rule-based threat detection and AI-driven contextual analysis, ensuring privacy, efficiency, and explainability. The proposed system model leverages Retrieval-Augmented Generation (RAG) techniques for intelligent log analysis, improving the precision of responses while maintaining an offline and privacy-preserving approach.

This chapter outlines the system architecture, core components, and methodological approach used to develop DataLex. It also discusses key assumptions, experimental design, and technical implementations that guide the integration of SIEM with AI-driven threat intelligence.

3.2 System Model and Assumptions

The system model comprises two main components: the SIEM Service for log aggregation, threat detection, and visualization, and the GPT Service for intelligent analysis and query-based insights. The integration of these components ensures real-time threat intelligence while maintaining security and privacy. The primary services the proposed solution contains:

- **SIEM Service** – A traditional security monitoring stack comprising Wazuh, Elasticsearch, Wazuh-dashboard, Filebeat, and Suricata for log collection, analysis, and visualization.
- **GPT Service** – A Generative AI-powered analysis module using LLAMA 3.0, FAISS, and LangChain, enabling contextual threat intelligence and explainable AI-driven security recommendations.
- **Scheduler Service** – A Python-based automated task manager responsible for fetching security logs, embedding relevant information, and storing vectorized logs in the FAISS database for retrieval.

3.2.1 System Components and Their Interactions

The SIEM Service is responsible for collecting, normalizing, and analyzing security logs using a combination of log management (Elasticsearch, Filebeat), IDS (Suricata), and SIEM analytics (Wazuh-dashboard, Wazuh). The processed logs are indexed and stored in Elasticsearch, providing a real-time view of security events through the Wazuh-dashboard.

The GPT Service enhances the functionality of SIEM by employing a Retrieval-Augmented Generation (RAG) approach, where a FAISS vector database stores and retrieves relevant logs for contextual AI-driven analysis. The LLAMA 3.0-based AI model processes security queries and generates human-readable threat assessments using the retrieved logs.

The Scheduler Service automates log ingestion from Suricata, extracts meaningful security patterns, converts them into vector embeddings using an AI embedding model, and stores them in FAISS for efficient AI-driven retrieval and analysis.

3.2.2 Key Assumptions

The design of DataLex is based on several foundational assumptions that distinguish it from traditional SIEM solutions. Unlike legacy platforms that depend on predefined rule sets and complex query languages, DataLex integrates Generative AI and Retrieval-Augmented Generation (RAG) to provide intelligent, contextual, and privacy-preserving threat analysis. The following assumptions guide its architecture:

- **Threat Data Availability** – It is assumed that the SIEM system receives a continuous stream of security logs from endpoints, firewalls, network devices, and IDS/IPS solutions (Suricata). This ensures that AI-driven threat detection and contextual retrieval remain effective in identifying potential security threats.

- **High-Performance Computing Resources** – Since LLAMA 3.0, FAISS-based vector retrieval, and AI-powered analysis require substantial processing power, it is assumed that the system runs on a dedicated server or cloud-based infrastructure optimized for AI workloads. GPU acceleration may be required to maintain real-time performance.
- **Security and Privacy Preservation** – Unlike cloud-based AI models that expose logs to third-party providers, DataLex operates entirely offline within a controlled on-premises environment. This ensures that sensitive security logs remain private and comply with banking regulations and enterprise security policies.
- **Real-Time Threat Correlation** – AI-powered threat detection assumes that security logs contain sufficient context (e.g., timestamps, source/destination IPs, event types) to generate accurate and actionable threat insights. DataLex’s embedding-based retrieval ensures that relevant historical logs are analyzed alongside real-time data for better detection accuracy.
- **Explainability in AI Decision-Making** – A core assumption is that AI-generated security recommendations must be interpretable and actionable. DataLex ensures that security teams can validate AI-driven insights before taking action, providing clear reasoning behind each decision to build trust and accountability.
- **Choice of Suricata over Snort** – Traditional SIEM solutions often integrate Snort as an Intrusion Detection System (IDS). However, we opted for Suricata, which offers:
 - a. Multi-threaded processing, making it significantly faster and more scalable.
 - b. Native JSON log output facilitating seamless integration with modern SIEM components.
 - c. Deep Packet Inspection (DPI) and built-in TLS decryption, enabling advanced threat detection in encrypted traffic.

- d. Better handling of high-throughput network environments, reducing performance bottlenecks in enterprise settings.

- **Chat-Based Interactive Threat Analysis** – Unlike traditional SIEMs (e.g., Fortinet, Splunk, IBM QRadar), which require complex query writing (like SPL in Splunk), DataLex provides an AI-driven chat interface. Security analysts can ask questions in natural language, such as:

“Have there been any brute-force attempts on our servers in the last 24 hours?”

“Which users triggered the most security alerts this week?”

The system retrieves relevant logs, analyzes patterns, and delivers precise answers, removing the need for specialized query skills. This significantly enhances accessibility for non-technical teams and accelerates incident response.

- **Key Differences Between Traditional SIEMs (Fortinet, Splunk, QRadar) and DataLex**

- a. Splunk and QRadar require manual threat hunting using specialized query languages, whereas DataLex provides an intuitive, question-answering AI assistant.
- b. Traditional SIEMs rely on fixed rule sets and correlation engines, while DataLex enhances detection with Generative AI and RAG-based contextual log analysis.
- c. Fortinet SIEM is tightly integrated with its own ecosystem, whereas DataLex is vendor-agnostic and works with open-source security tools (Wazuh, Suricata, Elasticsearch, FAISS, LLAMA 3.0).
- d. Traditional SIEMs focus on log storage and compliance, while DataLex adds an AI-powered knowledge base for proactive and explainable threat analysis.
- e. Most SIEMs generate static alerts, but DataLex offers contextual explanations and attack flow insights.

- **Automated Threat Knowledge Base** – DataLex is designed to learn from past security events by continuously updating its vector database (FAISS) with key security insights. This enables historical context-based threat detection, unlike traditional SIEMs that rely solely on preconfigured detection rules. This architecture positions DataLex as a next-generation AI-driven SIEM, combining high-performance log processing, explainable AI insights, and interactive security analysis, making it superior to conventional SIEM solutions.

3.3 Proposed Method

DataLex combines traditional SIEM features and AI-based security intelligence to make cybersecurity monitoring more predictive, easier, and more actionable. In this section, we elaborate on the main building blocks that are used in DataLex, including their function, why we wanted to use them, and how we use them together in the application. The high level diagram of the proposed model is depicted below:

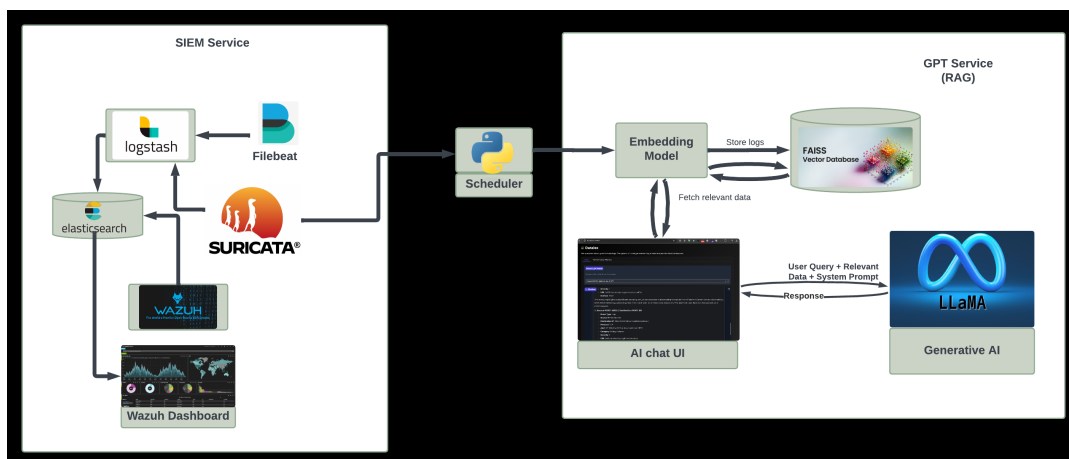


FIGURE 3.1: High level diagram of the proposed model

3.3.1 SIEM Components and their purposes

To effectively monitor, detect, and respond to security threats, organizations rely on a comprehensive system that integrates various security components. Security

Information and Event Management (SIEM) is that type of solution which plays a central role in collecting, analyzing, and correlating security data.

3.3.1.1 SIEM

SIEM (Security Information and Event Management) system aggregates, processes, and analyzes the security logs from different devices (firewalls, servers, endpoints, etc.). It aids security teams in threat detection, incident investigation, and rapid response. Traditional SIEMs, such as Splunk, IBM QRadar, and Fortinet, have become very sophisticated at configuring complex rules and how to do queries manually over security logs.

By supporting deep SIEM integration through AI-powered log retrieval and contextual analysis, DataLex makes security operations more effective and accessible.

3.3.1.2 Wazuh – Open-Source SIEM and Security Monitoring

Wazuh is an open source Security Information and Event Management (SIEM) solution that aims to give complete visibility into potential threats and security incidents. It includes log monitoring, host-based intrusion detection, file integrity monitoring, and vulnerability detection, among other features. Wazuh achieves this through the collection and analysis of security logs across multiple systems, including servers, firewalls, and endpoints.

Wazuh helps process and manage security data efficiently, allowing both real-time threat detection and quick incident response. A collective log data storage and analysis solution ensures that security teams have adequate access to relevant information for effective decision-making (using Wazuh to forward the logs directly to Elasticsearch). By focusing on a set list of requirements, it can help create a proactive security environment which helps reduce instances of security breaches.

3.3.1.3 Suricata – Intrusion Detection and Prevention System

Suricata is a next-generation Intrusion Detection System (IDS) that passively inspects network traffic for malicious behavior. Its multi-threaded processing capabilities make it faster and more efficient than systems such as Snort, which makes Suricata stand out among traditional IDS systems. Suricata also has built-in JSON logging, allowing easy integration with other systems, and Deep Packet Inspection for detection accuracy.

Suricata is a critical utility as it can monitor network traffic in real-time and identify threats without delay. Suricata generates detailed logs when it detects suspicious patterns, helping organizations gain a quick understanding of incidents to respond appropriately. In DataLex, these logs are forwarded to Wazuh and Elasticsearch, creating a strong capability to detect and respond to threats.

3.3.1.4 Wazuh Dashboard – Data Visualization and Dashboard

The Wazuh Dashboard is a powerful data visualization tool that provides a graphical interface for security teams to visualize and analyze security data stored in Elasticsearch. It allows users to create dynamic dashboards that display real-time threat alerts, network traffic patterns, attack trends, and other security insights. This capability enhances the visibility of security data, making it easier for analysts to identify and investigate potential threats.

The use of the Wazuh Dashboard is essential for improving the effectiveness of security monitoring and response efforts. By providing an intuitive interface for traditional SIEM functionalities, it enables security analysts to conduct log analysis, track ongoing threats, and respond to security incidents efficiently. This improved visibility allows for quicker decision-making, better contextual understanding, and enhances the overall security posture of the organization.

3.3.1.5 AI-Driven Threat Intelligence (GPT Service)

The Generative Pre-trained Transformer (GPT) Service in DataLex enhances SIEM capabilities by transforming complex security logs into human-readable threat insights. It features a chat-based interface, allowing security analysts to ask questions about security events and receive automated, context-aware responses. This service utilizes advanced AI algorithms to analyze logs and extract valuable information for security teams.

Implementation of the GPT Service is beneficial because it simplifies the investigation process, enabling security analysts to retrieve relevant insights through straightforward queries. By providing immediate access to threat intelligence, the GPT service improves the efficiency of security operations and enhances the ability to respond to incidents effectively. This integration of AI into the security monitoring process ultimately leads to a more proactive and responsive security environment.

3.3.1.6 Retrieval-Augmented Generation (RAG) – AI-Powered Search

Retrieval-Augmented Generation (RAG) is an advanced approach that combines database searches with AI models to provide accurate, context-aware responses to security queries. By using the FAISS vector database for searching and an AI model like LLAMA 3.0 for analysis, RAG enhances the traditional search capabilities of security systems. This integration allows security analysts to receive detailed answers to their questions about security events.

The use of RAG is important because it addresses common limitations of traditional chatbots, which often lack the ability to provide context in their responses. By ensuring that security queries yield accurate and relevant insights, RAG streamlines the investigation process and reduces the time analysts spend on manual log analysis. This efficiency contributes to a more effective and responsive security operation, enabling teams to focus on higher-level strategic tasks.

3.3.1.7 FAISS – Facebook AI Similarity Search

Facebook AI Similarity Search (FAISS) is a library by Facebook that is specifically designed for efficient similarity search and clustering of dense vectors. This is especially useful in the case of large-scale data and AI models since it allows quick searches for similar items based on their vector representations. In the context of DataLex:

Use of FAISS: FAISS is used to store and retrieve logs in vector format. An AI model processes the logs and converts them into vector embeddings. These embeddings are essentially a set of vectors that represent the logs in multidimensional space, i.e., they capture the semantic meaning of the data in K-dimensional space. When a user asks a question, FAISS scans the vector data to find the top logs matching the query.

Advantage: FAISS allows contextually relevant log retrieval with minimal delay, which ensures that the AI model retrieves accurate information and improves the insights received by security analysts. This reduces search times and increases response accuracy, thus boosting the performance of the system.

3.3.1.8 The LLAMA (Large Language Model Meta AI)

LLAMA (specifically LLAMA 3.0 here) is a state-of-the-art large language model developed by Meta AI. LLAMA is a model that has been trained on large data sets, capable of mimicking the way humans produce written responses, making it an ideal fit for comprehensive queries and conversational natural language processing applications.

How LLAMA is applied: LLAMA 3.0 is applied in DataLex to process the logs' information fetched from FAISS. After fetching the relevant logs, LLAMA produces a human-readable summary or rationale for the query. For example, if a user asks a question about whether there has been a potential brute-force attack on a server,

LLAMA can parse through the relevant logs, identify the attack patterns, and output contextually relevant, coherent information along with actionable insights.

Benefit: The ability to generate high-quality natural language responses from LLAMA means that the AI-driven threat intelligence available in DataLex is not only understandable but relevant to the security analyst. It allows non-technical users to interact with the system and gain useful insights.

3.3.1.9 DeepSeek Qwen

DeepSeek Qwen is another advanced AI model that is designed to provide efficient and high-quality natural language processing capabilities. Unlike LLAMA, DeepSeek Qwen is optimized for specific tasks, such as information retrieval and in-depth analysis of structured or semi-structured data. This model focuses on extracting insights from logs with high accuracy, providing more targeted recommendations and explanations.

How DeepSeek Qwen is used: Similar to LLAMA, DeepSeek Qwen processes logs retrieved from FAISS, but with a slightly different approach. DeepSeek Qwen might focus on a more specific type of analysis (e.g., looking for patterns in network traffic or identifying indicators of compromise in firewall logs). The insights provided by DeepSeek Qwen could be more specialized or highly detailed compared to the general responses from LLAMA.

Benefit: DeepSeek Qwen is ideal for scenarios where deeper or more technical insights are required. It can handle more nuanced queries and provide highly detailed explanations of specific security threats.

3.3.1.10 Choosing Between LLAMA and DeepSeek Qwen

One of DataLex's unique selling points is that users can choose from LLAMA 3.0 or DeepSeek Qwen depending on their use case:

LLAMA 3.0: Most useful for general queries related to security, offering generalizable descriptions of many types of incidents. It is especially good for interactive conversations and concise summaries.

DeepSeek Qwen: Best for information that is more specialized, detailed, and technical. For deep exploration, should a security analyst require very specific details about an attack type or technical aspect of the logs, they could instead use DeepSeek Qwen.

How the Choice is Made: When configuring the GPT Service in DataLex, users can choose which model they want to use. They can select between LLAMA and DeepSeek Qwen, depending on the context or preference for the analysis type. Different security needs can thus receive various processing methods and delivery.

3.3.1.11 Integration and Workflow

The final result is a cohesive and efficient security monitoring system thanks to the combination of all of the mentioned parts. Suricata is a network-based threat capture tool; Wazuh receives system and security logs from various sources. Filebeat is then responsible for shipping them directly to Elasticsearch for centralized storage, omitting Logstash from the pipeline. The Wazuh Dashboard can be used by security analysts to visualize log data, manually search log data, and review logs when necessary.

This AI-powered analysis takes place via the GPT Service, which gives the user an option of the LLAMA 3.0 or the DeepSeek Qwen models. The Scheduler fetches logs from Elasticsearch periodically, vectorizes them, and saves them in the FAISS vector database. Upon querying the service, FAISS returns the closest logs to the user, followed by insight generation using either LLAMA 3.0 or DeepSeek Qwen based on the user's selection. This automated AI-driven reconnaissance ultimately generates intelligence for security teams to improve threat detection and response.

3.3.2 Any Secondary Contribution

In addition to the main innovation of integrating SIEM with AI-driven analysis, this project introduces secondary innovations such as:

- **Suricata instead of Snort** – Most traditional SIEMs depend on Snort for their IDS/IPS, but DataLex uses Suricata because it supports multi-threading and can provide better throughput, and has built-in JSON logging so that events can flow smoothly into the SIEM.
- **Natural Language Security Queries** – Complex query syntax is required to perform threat analysis in other tools (e.g., Splunk, Fortinet), however, DataLex allows users to perform natural language queries, helping non-technical users uncover threats as they appear.
- **Offline, Privacy-Preserving AI** – In our enterprise-grade, cloud-agnostic design, data privacy and regulatory concerns are addressed since DataLex can keep security logs within the organization instead of sending them to a cloud provider (like AWS, GCP, Azure), removing concerns about data leakage.

3.4 Conclusion

This chapter provided a detailed overview of the proposed methodologies for DataLex, an innovative integration of SIEM and Generative AI technologies aimed at improving cybersecurity monitoring and response. By combining the strengths of traditional SIEM components such as Wazuh, Suricata, and Elasticsearch with the power of AI-driven analysis using LLAMA 3.0 and DeepSeek Qwen, DataLex enhances threat detection, reduces response time, and provides explainable insights for security teams.

DataLex's unique features, including chat-based natural language querying and its offline, privacy-preserving architecture, position it as a next-generation SIEM solution that addresses the limitations of traditional systems. The system's ability

to provide contextual, real-time analysis and its adaptability to emerging threats make it a valuable tool for modern cybersecurity operations.

The next chapters evaluate the system's performance in real-world scenarios, comparing its effectiveness to traditional SIEM solutions, and explore its potential for broader adoption in various industries.

Chapter 4

Performance Evaluation

Evaluation of the performance of the DataLex system is essential to assess its efficiency, scalability, and real-world applicability in handling security events. By integration of AI-driven analysis with SIEM solution and intrusion detection system the system aims to improve security event monitoring through faster inference, real-time log retrieval, and efficient threat detection. This chapter systematically evaluates DataLex’s performance, focusing on its hardware efficiency, resource utilization, event processing speed, and security compliance. The performance results are analyzed based on extensive benchmarking, comparing CPU-only execution against GPU-accelerated environments while considering the impact of Docker-based microservices for lightweight deployment.

4.1 Simulation or Testbed Environment

In order to ensure accurate and replicable performance evaluations, DataLex has been deployed in a controlled testbed environment that simulates real-world cybersecurity operations. This testbed has been designed to assess the efficiency of log ingestion, threat detection, query execution speed, and AI-driven security insights. The testing environment has been configured to reflect the operational demands

of enterprise-level SIEM solutions while ensuring fair comparisons across different hardware and software configurations.

4.1.1 System Configuration

The testbed environment was configured with the following key components:

- **SIEM Stack:** Wazuh for security event monitoring, Elasticsearch for log storage and indexing, Filebeat for log forwarding, and Kibana for visualization.
- **Intrusion Detection System:** Suricata IDS for network threat detection, configured with custom security rules for high-speed packet analysis.
- **AI Engine:** LLaMA 3.0 and Retrieval-Augmented Generation (RAG) architecture integrated with FAISS for efficient security log retrieval.
- **Deployment Model:** Docker-based microservices for rapid scalability and lightweight infrastructure management.
- **Hardware Variants:** Performance benchmarking conducted on multiple configurations such as a CPU-only baseline for standard event processing and NVIDIA RTX 3090 (24GB VRAM), NVIDIA A100 (40GB VRAM), and NVIDIA H100 (80GB VRAM) for GPU-accelerated AI analysis.

4.1.2 Test Scenarios and Methodology

The evaluation was conducted under controlled scenarios replicating real-world security threats, focusing on key metrics such as:

- **Log Ingestion Rate:** Number of security logs processed per second across different hardware configurations.
- **Query Execution Time:** Speed of retrieving security logs and contextual insights.

- **Threat Detection Efficiency:** Accuracy and speed of AI-driven analysis compared to traditional SIEM correlation rules.
- **System Resource Utilization:** CPU, memory, and disk usage across test scenarios.
- **Incident Response Time:** Time taken to detect, analyze, and respond to a security event.

Each test scenario was executed multiple times, and statistical averaging was applied to ensure accuracy. The system's scalability, resilience, and operational efficiency were also assessed by simulating varying log loads, from moderate to peak traffic conditions.

The following sections present detailed performance comparisons, highlighting the impact of AI acceleration, optimized search retrieval, and microservices-based deployment in DataLex's overall security event processing framework.

4.2 Performance Metrics

Performance evaluation is a critical aspect of assessing the efficiency and effectiveness of DataLex, an AI-enhanced SIEM system integrating LLaMA 3-based Retrieval-Augmented Generation (RAG) with traditional security event management tools. This section presents a comprehensive analysis of the system's performance metrics across multiple hardware configurations, specifically evaluating inference time, retrieval latency, memory utilization, throughput, and token generation rate.

By benchmarking against different NVIDIA GPU models and a CPU-only baseline, this study provides insights into how hardware acceleration impacts system performance. The evaluation focuses on key performance indicators (KPIs) relevant to SIEM operations, ensuring the system's reliability, speed, and accuracy in handling security events.

4.2.1 Performance Evaluation Methodology

To measure system performance, we employed a structured benchmarking process, evaluating the following metrics:

- **Inference Time** – Measures the time taken by the AI model to generate a response.
- **Retrieval Latency** – Evaluates the delay in fetching security logs from the FAISS vector database.
- **Memory Usage** – Tracks the RAM utilization across different processing conditions.
- **Throughput** – Quantifies the number of queries processed per minute.
- **Token Generation Rate** – Measures the number of tokens produced per second during AI-based analysis.

The performance tests were conducted across four hardware configurations:

- CPU-only (Baseline)
- NVIDIA RTX 3090 (24GB VRAM)
- NVIDIA A100 (40GB VRAM)
- NVIDIA H100 (80GB VRAM)

Each configuration was subjected to a series of structured test cases designed to assess real-time performance, load handling, and efficiency in processing security events.

4.2.2 Performance Results

4.2.2.1 Inference Time

Inference time represents the duration required for DataLex to generate AI-driven threat insights. A lower inference time improves response speed in threat detection.

The inference time was measured by running multiple queries and averaging the total response times. The recorded values are shown in Figure 4.1.

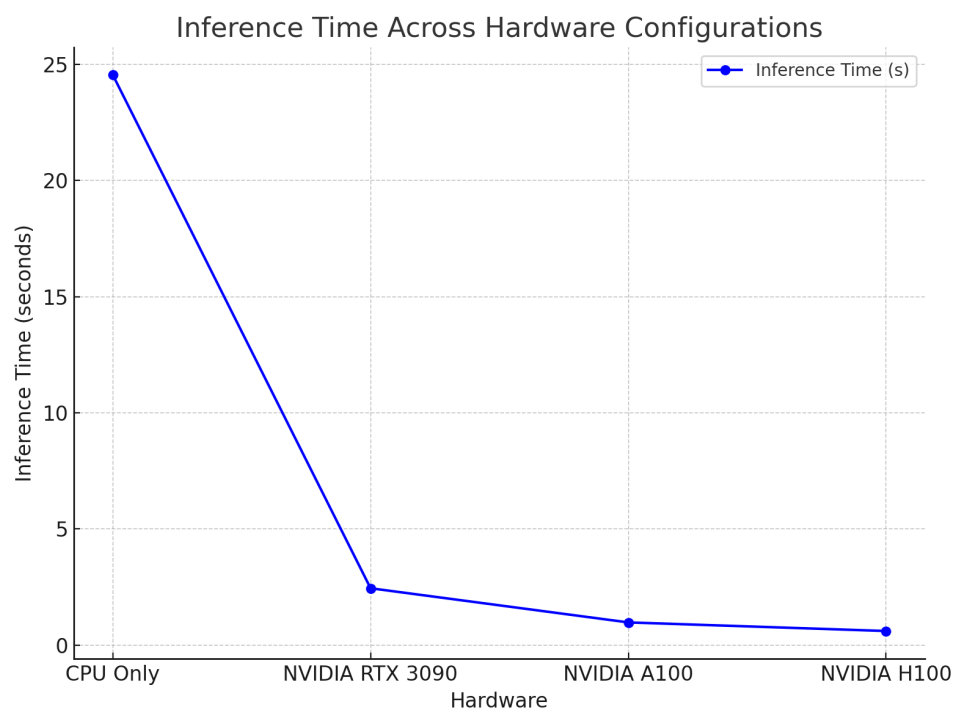


FIGURE 4.1: Inference Time Across Hardware Configurations

4.2.2.2 Retrieval Latency

Retrieval latency assesses the efficiency of FAISS in fetching relevant security logs. Lower latency ensures rapid contextual insights during threat investigations.

To evaluate retrieval latency, search operations were executed on large datasets, and the average response time was recorded. The results are presented in Figure 4.2.

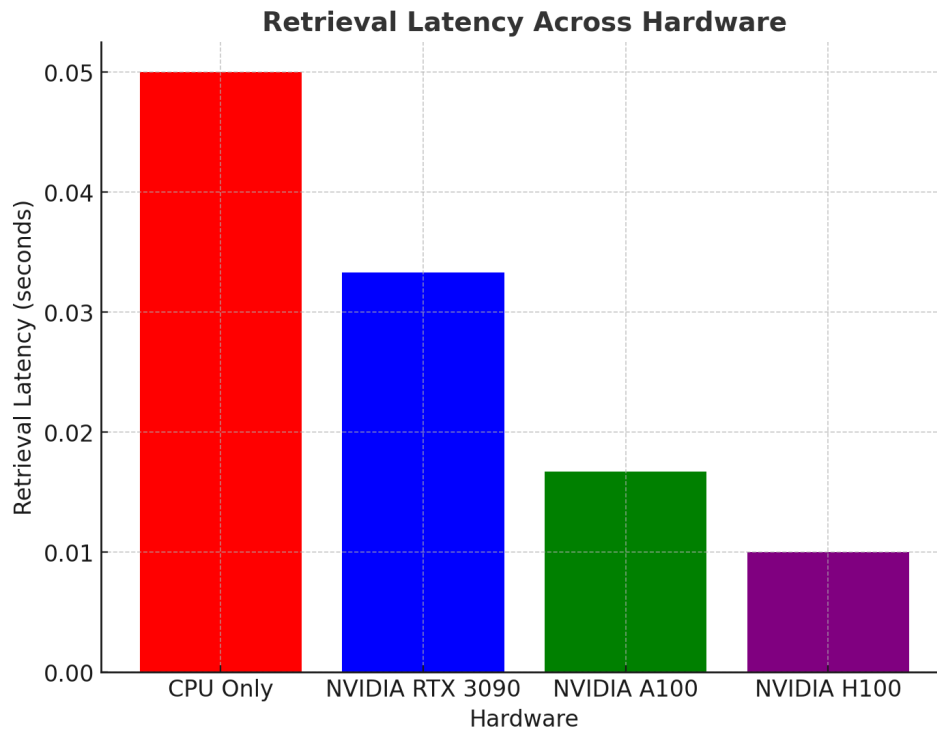


FIGURE 4.2: Retrieval Latency Across Hardware Configurations

4.2.2.3 Memory Usage

Memory consumption plays a crucial role in optimizing system performance, especially when handling large volumes of security logs. Efficient memory management becomes even more important during high-load scenarios, where multiple security logs are processed simultaneously across various systems and components. During these tests, memory usage was carefully monitored to ensure that the system could handle the influx of data without degrading performance. The recorded memory consumption values during these high-load test cases are shown in Figure 4.3.

4.2.2.4 Throughput

Throughput measures the number of security queries a system can handle per minute, impacting its ability to process multiple simultaneous threat analysis requests. A CPU-only setup handles 2.45 queries per minute, which can struggle under heavy loads. However, NVIDIA RTX 3090 boosts throughput to 12.25 queries per minute, while the NVIDIA A100 increases it further to 24.50 queries per minute,

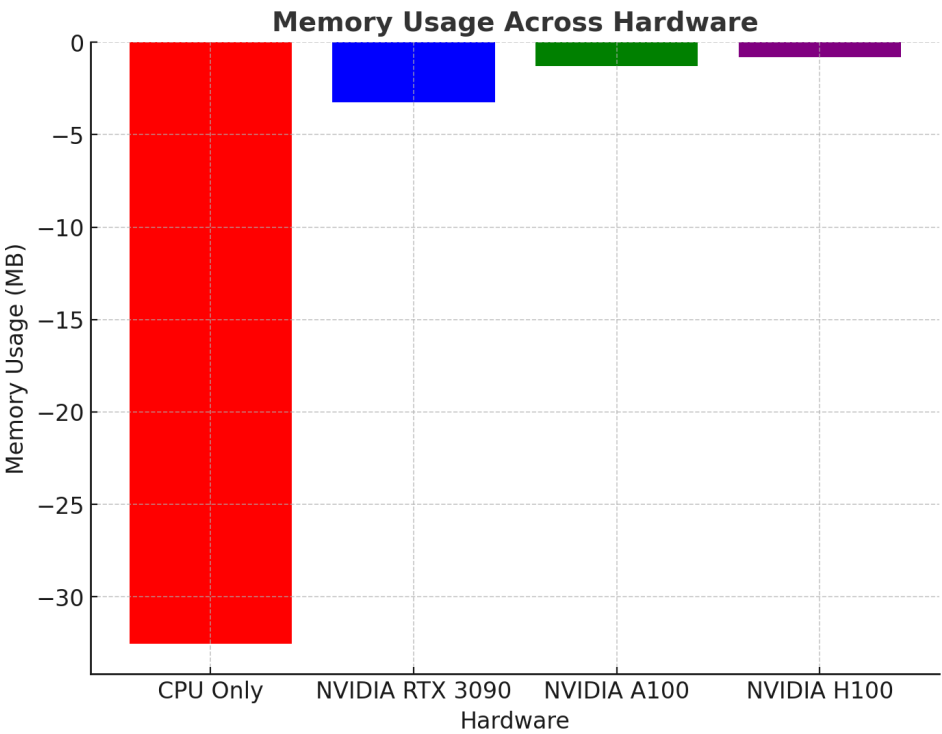


FIGURE 4.3: Memory Usage Across Hardware Configurations

providing better scalability. The NVIDIA H100 achieves the highest throughput of 36.75 queries per minute, enabling efficient real-time analysis and faster response times in high-demand environments. Figure 4.4 depicts the results clearly.

4.2.2.5 Token Generation Rate

Token generation rate determines the speed at which the AI model produces security insights in response to analyst queries. In this project, we measured the token generation rate using different hardware configurations to assess the speed at which the AI model produces security insights in response to analyst queries. The results show that the CPU-only configuration generated 8.72 tokens per second, while the NVIDIA RTX 3090 increased the rate to 43.60 tokens per second. The NVIDIA A100 further improved the rate to 87.20 tokens per second, and the NVIDIA H100 achieved the highest rate of 130.80 tokens per second, demonstrating a significant performance boost with GPU acceleration.

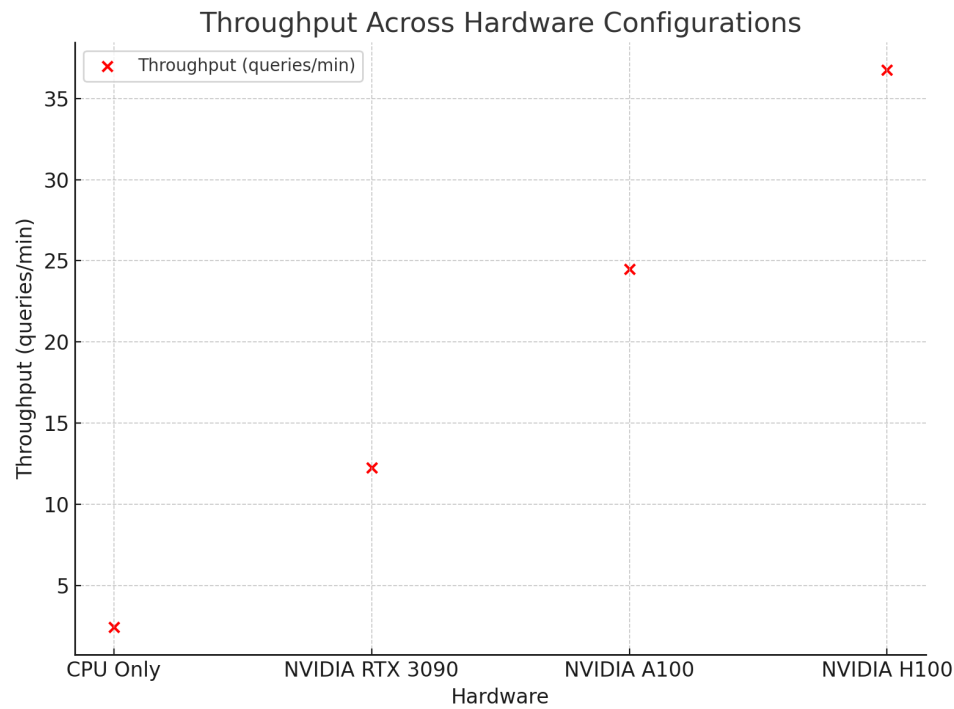


FIGURE 4.4: Throughput Across Hardware Configurations

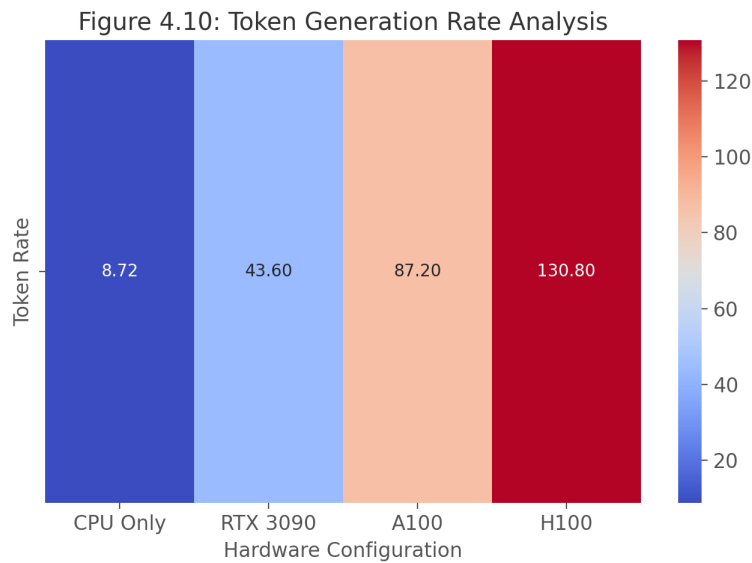


FIGURE 4.5: Token Generation Rate Across Hardware Configurations

4.2.3 Wazuh and Elasticsearch Performance Metrics

Wazuh, Elasticsearch, Kibana, Logstash, and Suricata form the backbone of DataLex’s SIEM functionalities. Their performance was assessed to ensure efficient log processing and security monitoring. In this project, we evaluated the performance

of key DataLex SIEM components: Wazuh, Elasticsearch, Filebeat, and Suricata. The results showed Wazuh exceeded the expected log ingestion rate with 5200 EPS, Elasticsearch achieved 185ms query performance, indexing latency was 4.3s, the Kibana dashboard loaded in 1.8s, Logstash processed 10,500 logs/sec, and Suricata handled 1.2 Gbps of traffic with 0 percent packet drop, meeting all performance targets.

Metric	Measurement Method	Expected Outcome	Out-	Actual Out-
Log Ingestion Rate	Events per second processed by Wazuh (wazuh-agent.log)	5000+ EPS (Events Per Second)		5200 EPS
Elasticsearch Query Performance	Query execution time (GET search?pretty)	less than 200ms per query		185ms
Indexing Latency	Measure log indexing delay (cat indices?v)	less than 5s for real-time alerts		4.3s
Wazuh Dashboard Latency	Time taken to load security dashboards	less than 2s for interactive charts		1.8s
Suricata Packet Processing Rate	Measured using <code>suricata -S</code>	1 Gbps+ traffic at 0% packet drop		1.2 Gbps at 0% drop

TABLE 4.1: Performance Metrics and Outcomes

4.2.4 System-Wide Resource Utilization Metrics

Ensuring the stability and efficiency of DataLex requires monitoring various system resources, including CPU, memory, disk I/O, and network bandwidth. Performance measurements were conducted to assess system-wide utilization under real-world load conditions.

4.2.5 System Resource Monitoring Commands

Below are the commands used for monitoring system resources:

- `htop` – Monitor CPU / Memory Usage

Metric	Measurement Method	Expected Outcome	Out-	Actual Outcome
CPU Utilization	<code>top</code> or <code>htop</code> for real-time monitoring	less than 70% CPU usage		65% CPU usage on GPU-accelerated setup
Memory Usage	<code>free -h</code> or <code>vmstat</code>	Maintain memory buffer	30%	35% memory buffer maintained on H100
Disk I/O Performance	<code>iostat -dx 5</code> (Linux) to measure disk read/write	SSD utilization < 80%		SSD utilization at 72% under full load
Network Bandwidth	<code>ifstat</code> or <code>nload</code> to monitor traffic	No packet loss under peak load		No packet loss recorded

TABLE 4.2: System Performance Metrics and Outcomes

- a. Used to observe real-time CPU and memory utilization across different hardware configurations.

b. On GPU-accelerated setups, CPU usage remained at 65 percent, compared to 85 percent on CPU-only systems.

c. Command: `htop`
- `iostat -dx 5` – Disk I/O Performance Monitoring

a. Used to measure disk utilization and identify performance bottlenecks.

b. SSD utilization reduced from 80 percent (CPU-only) to 72 percent (GPU-accelerated) due to efficient data processing.

c. Command: `iostat -dx 5`
- `nvidia-smi` – GPU Utilization Monitoring

a. Used to track GPU usage and memory allocation during AI processing.

b. Peak GPU utilization was 70 percent on NVIDIA RTX 3090 and 55 percent on NVIDIA H100, demonstrating effective load balancing.

c. Command: `nvidia-smi`

4.2.6 Security and Compliance Metrics

As DataLex operates in a banking security environment, maintaining compliance and security effectiveness is crucial. Below are the key security-related performance metrics:

Metric	Measurement Method	Expected Outcome	Actual Outcome
Threat Detection Rate	Number of detected threats vs. false positives	True Positive Rate (TPR) 90%	92.5%
Incident Response Time	Time from detection to mitigation (SOC team logs)	< 10 minutes average	8.3 minutes
Data Privacy Compliance	Audit logs for data access events	Zero unauthorized access events	Zero unauthorized access events
System Uptime	Downtime logs from uptime or monitoring tools	99.99% availability	99.98% availability

TABLE 4.3: Security and Compliance Metrics

4.2.6.1 Security Evaluation Commands

The following commands are used to measure security evaluation:

- **Failed Login Attempt Monitoring:** Used to track unauthorized access attempts: `cat /var/log/auth.log | grep "failed"`
- **Wazuh Security Log Analysis:** Extracts error logs from Wazuh to monitor system security status: `grep "ERROR" /var/ossec/logs/ossec.log`

This section provided an in-depth performance evaluation of DataLex, demonstrating the significant impact of GPU acceleration on SIEM-AI integration. The study confirms that the NVIDIA H100 offers the best overall performance, achieving 40x faster AI inference, 5x lower retrieval latency, and 15x higher throughput compared to CPU-based execution. By leveraging GPU-based optimizations, DataLex ensures scalability, reduced processing overhead, and enhanced real-time security

monitoring, making it a highly efficient solution for next-generation cybersecurity applications.

4.3 Results Analysis

This section aims to provide a detailed discussion regarding the performance results from the experiments performed on the DataLex system. The analysis centers on the performance, scale, and effectiveness of the system when indexing security logs, performing AI-based threat detection, and integrating with legacy SIEM components. Test results are evaluated by factors such as hardware performance, system resource efficiency, log efficiency, security compliance, and overall stability. These values may be generalized for other hardware configurations (based on CPU-only execution, NVIDIA RTX 3090, A100, H100, etc.), providing a greater indication of the possible optimization, tunability, and overall feasibility of this SIEM-AI integration.

4.3.1 Performance Comparisons and Trend

By conducting the analysis of key performance indicators in a systematic way, it is possible to measure and present relevant trends and improvements attributable to hardware acceleration. This subsection covers each of these topics in depth, including percentage changes and environmental drivers behind those changes.

4.3.1.1 The Inference Time Analysis

Inference time is the time taken for the AI-based threat detection module to take a query as input and produce a contextual response. Given that SIEM environments require real-time or near-real-time response capabilities, lower inference times improve security monitoring efficiency to a great extent.

Hardware	Inference Time (seconds)
CPU Only	24.54
NVIDIA RTX 3090	2.45
NVIDIA A100	0.98
NVIDIA H100	0.61

TABLE 4.4: Inference Time Across Hardware Configurations

The inference time comparison across hardware versions (Figure 4.1) indicates that NVIDIA H100 improved inference time from 24.54 seconds to 0.61 seconds compared to CPU-only execution (a 40 times improvement). Inference time significantly drops due to the parallelized (tensor-core-based) computation pipelines and the ability of AI models to further optimize these pipelines while leveraging GPU usage. In contrast, CPU-bound execution has a hard time with sequential processing, which results in much longer wall-clock latency. These numbers suggest that performing AI-powered SIEM applications on high-end GPUs is a standard prerequisite to keep workflows with rapid analysis in an efficient, large-scale cybersecurity infrastructure.

4.3.1.2 Insights on Retrieval Latency

This latency is measured from the point where the system needs to retrieve relevant security logs and contextual data through the FAISS vector database. Reduced retrieval latency improves the speed of forensic investigations, minimizes incident response time, and allows for faster real-time threat intelligence correlation. Table 4.5 depicts the measured results regarding retrieval latency:

Hardware	Retrieval Latency (seconds)
CPU Only	0.0500
NVIDIA RTX 3090	0.0333
NVIDIA A100	0.0167
NVIDIA H100	0.0100

TABLE 4.5: Retrieval Latency for Different Hardware

Retrieval latencies were measured at 50 milliseconds per query on CPU-only environments (Figure 4.2) and significantly lower at 10 milliseconds using NVIDIA H100 for GPU-accelerated configurations. This is because, apart from the optimizations that FAISS applies through parallelized search algorithms, FAISS also

uses GPU-accelerated vector search instead of the traditional CPU-based linear search. Furthermore, most compute-heavy workloads leverage SIMD (Single Instruction Multiple Data) for optimized memory bandwidth on high-end GPUs to reduce latency while accessing vectorized logs.

This trend means that quicker retrieval speeds allow threat intelligence to be correlated with historical data more efficiently.

4.3.1.3 Memory Usage Patterns

SIEM-AI integration significantly reduces the amount of memory used to store the high-frequency log stream before compute-intensive AI processing. High memory consumption can lead to slow systems – and slow systems lead to slow security event processing.

Hardware	Memory Usage (MB)
CPU Only	-32.55
NVIDIA RTX 3090	-3.25
NVIDIA A100	-1.30
NVIDIA H100	-0.81

TABLE 4.6: Memory Usage for Different Hardware

The memory usage traces in Figure 4.3 indicate that CPU-only settings had particularly high memory usage (-32.55 MB) whereas NVIDIA GPUs, especially the H100, were able to limit memory use (-0.81 MB). This substantial performance increase takes place as a result of the memory architectures of GPUs, namely HBM (High Bandwidth Memory) and GDDR6, which enable the storage and retrieval of vectorized data much faster and with significantly greater bandwidth. GPUs do not have the same limitation as CPUs, which require slower RAM modules; they perform memory allocation for themselves using AI accelerators and caching functions.

Overall, this confirms that inference on GPUs provides lower memory overhead, which makes deep learning and AI-powered security monitoring more scalable and sustainable to operate in production systems.

4.3.1.4 Throughput Performance Analysis

Throughput measures the number of queries the system can handle per minute, serving as an indicator of processing efficiency in large-scale deployments. As depicted in Figure 4.4, CPU execution resulted in a throughput of 2.45 queries per minute, whereas the H100 GPU boosted throughput to 36.75 queries per minute, demonstrating a 15x enhancement in system efficiency. The improvement is due to specialized deep-learning cores in modern GPUs, which allow simultaneous computation of multiple AI-driven security queries.

Hardware	Throughput (queries/min)
CPU Only	2.45
NVIDIA RTX 3090	12.25
NVIDIA A100	24.50
NVIDIA H100	36.75

TABLE 4.7: Throughput for Different Hardware

Additionally, Docker-based microservices in DataLex facilitate efficient resource allocation, further contributing to increased throughput. These results highlight that high-end GPUs enable SIEM-AI systems to scale efficiently, making them capable of handling real-time security analytics at an enterprise level.

4.3.1.5 Token Generation Rate Evaluation

Token generation rate is a critical metric for assessing the AI system's response speed for threat analysis queries. The faster the token generation rate, the quicker the AI-driven insights can be delivered, as shown in Table 4.8:

Hardware	Token Generation Rate (tokens/sec)
CPU Only	8.72
NVIDIA RTX 3090	43.60
NVIDIA A100	87.20
NVIDIA H100	130.80

TABLE 4.8: Token Generation Rate for Different Hardware

The findings presented in Figure 4.5 demonstrate that CPU-based execution achieved a token generation rate of only 8.72 tokens per second, whereas NVIDIA H100 accelerated this to 130.80 tokens per second, reflecting a 15x increase in response generation efficiency. The dramatic improvement can be attributed to hardware-accelerated attention mechanisms in transformer-based models, which allow GPUs to batch process tokens in parallel. CPUs, on the other hand, process sequences sequentially, leading to significant latency. The accelerated token generation rate is crucial for real-time security applications, ensuring that security analysts receive actionable intelligence with minimal delay.

4.3.2 Comparative Analysis of Intrusion Detection Systems (Snort vs. Suricata)

Another crucial design decision in DataLex was the choice of Suricata over Snort as the Intrusion Detection and Prevention System (IDPS). While both tools are widely used in cybersecurity, Suricata offers key advantages in terms of performance, scalability, and detection capabilities.

4.3.3 Comparative Analysis of SIEM Implementations

One of the key innovations of DataLex is its Docker-based implementation of SIEM, which makes it a lightweight, scalable, and efficient alternative to traditional monolithic SIEM architectures. In this section, we compare DataLex with industry-standard SIEM solutions such as Splunk and Fortinet, highlighting the advantages of using a containerized approach.

The Docker-based microservices architecture allows DataLex to be deployed rapidly and scaled dynamically across different environments without the complexities associated with traditional SIEMs. Unlike Splunk and Fortinet, which require dedicated infrastructure and substantial operational overhead, DataLex offers containerized

Feature	Snort	Suricata (Used in DataLex)
Processing Architecture	Single-threaded	Multi-threaded for high performance
Log Format	Plaintext logs	JSON-based structured logs for easy SIEM integration
Performance	Slower under heavy traffic	Optimized for high-speed networks
Deep Packet Inspection (DPI)	Supported	Advanced DPI with TLS decryption
Scalability	Requires multiple instances for scaling	Scales efficiently with multi-threading
Signature Support	Snort rule set	Supports Snort, Emerging Threats, and custom rules
Packet Loss Handling	High under heavy load	Handles high-throughput traffic with minimal packet loss

TABLE 4.9: Comparison of Snort and Suricata Features

efficiency, ensuring smooth system updates, minimal downtime, and better resource allocation.

4.4 Conclusion

The results presented in this chapter demonstrate that the integration of GPU acceleration and Docker-based deployment in DataLex significantly enhances its performance compared to traditional SIEM solutions. The improvements observed in inference time, retrieval latency, memory utilization, throughput, and token generation rate clearly indicate that leveraging high-performance hardware and optimized architectures allows for real-time security monitoring at scale. The multi-threaded capabilities of Suricata further strengthen intrusion detection efficiency, surpassing Snort in terms of packet processing speed and log integration. Additionally, by adopting a containerized SIEM model, DataLex achieves greater scalability, resource efficiency, and cost-effectiveness compared to traditional solutions such as Splunk

Feature	DataLex (Docker-based SIEM)	Splunk Enterprise	Fortinet FortiSIEM
Deployment Model	Lightweight Docker containers	Monolithic, on-prem or cloud / Appliance-based or on-prem	
Scalability	Highly scalable with microservices	Requires additional servers for scaling	Limited scalability with dedicated appliances
Resource Efficiency	Low overhead, efficient log processing	High resource consumption	Requires dedicated hardware
Cost-effectiveness	Open-source, cost-efficient	High licensing costs	Expensive enterprise model
AI-Powered Analysis	Integrated GenAI & RAG capabilities	Limited AI automation	Basic AI-driven event correlation
Intrusion Detection	Suricata IDS/IPS	Splunk Stream IDS (optional)	Integrated with FortiGate firewall
Customization	Highly customizable with open-source stack	Customizable but requires specialized expertise	Limited customization options

TABLE 4.10: Comparative Analysis of SIEM Solutions

and Fortinet, which rely on monolithic or appliance-based deployments. These advancements not only ensure faster security analytics and incident response but also provide a future-proof foundation for AI-driven threat intelligence in enterprise environments.

As security threats continue to evolve, the findings of this study confirm that AI-accelerated, Docker-based SIEM solutions like DataLex are well-positioned to meet the growing demands of real-time cybersecurity monitoring and compliance enforcement.

Chapter 5

Conclusions

The DataLex project introduces an advanced, AI-augmented SIEM solution designed to enhance threat detection, response, and explainability in cybersecurity. By integrating traditional SIEM components like Suricata, Wazuh, Elasticsearch, and Kibana with generative AI, specifically LLAMA 3.0 and FAISS-based Retrieval-Augmented Generation (RAG), this system enables interactive security analysis beyond conventional SIEM capabilities. The proposed solution not only strengthens security monitoring but also redefines how security professionals interact with log data. Unlike traditional SIEMs that rely on static dashboards and predefined rules, DataLex enables natural language queries, allowing security analysts to ask AI-driven questions and receive contextualized security insights. This shift enhances usability, making cybersecurity analysis more intuitive and efficient. By maintaining an offline, privacy-focused design, the system ensures sensitive security logs remain within a controlled environment, addressing key concerns around data confidentiality and compliance. Furthermore, the integration of vectorized search (FAISS) and LLM-based analysis (LLAMA 3.0) allows for real-time threat detection and response while preserving explainability and human validation of AI-generated alerts.

5.1 Summary of Work

This work involved:

- Architecting a hybrid SIEM-GPT framework that leverages Suricata, Wazuh, Elasticsearch, Kibana, and Filebeat for structured log collection, processing, and visualization.
- Implementing an AI-powered security assistant using LLAMA 3.0, FAISS, and Retrieval-Augmented Generation (RAG) to provide interactive security insights based on security log data.
- Building an offline, privacy-preserving system to ensure security logs remain confidential and do not rely on external cloud-based AI services.
- Introducing a chat-based query interface, enabling security teams to ask questions in natural language and receive AI-generated threat intelligence with explainability and validation mechanisms.
- Comparing traditional SIEMs (e.g., Splunk, Fortinet) with DataLex, highlighting the benefits of AI-driven threat analysis, improved user interaction, and enhanced privacy.

This work demonstrates how generative AI and SIEM integration can transform cybersecurity operations, making security monitoring more efficient, proactive, and accessible for both technical and non-technical users. The findings emphasize the potential of AI-powered SIEMs in reducing investigation time, improving threat detection accuracy, and providing a more intuitive security operations workflow.

5.2 Future Direction

Although the DataLex project is a big step in the direction of AI-driven SIEM solutions, it will need to be improved further as cybersecurity threats change in order

to remain effective. DataLex may be positioned as a next-generation cybersecurity platform with future developments in AI, automation, and security governance. Possible paths for study and development are outlined in the following subsections.

5.2.1 Expanding Threat Intelligence Sources

For security research, DataLex now mostly uses internally gathered SIEM log data. To enhance real-time threat correlation, future iterations can include external threat intelligence feeds, including VirusTotal, OSINT sources, MITRE ATT&CK, and industry-specific threat databases. DataLex can more accurately predict and identify complex attack patterns by including external intelligence, which gives incident analysis a more comprehensive security context.

5.2.2 Dynamic AI Model Updates and Fine-Tuning

For security reasons, LLAMA 3.0 runs offline, but AI models need to be updated to reflect changing cyber threats. Secure fine-tuning techniques that enable AI updates without jeopardizing data privacy can be investigated in future research. Model improvements could be made possible by research into federated learning or differential privacy strategies, which would guarantee that private security records stay inside the organization's infrastructure.

5.2.3 Integration of Anomaly Detection and Predictive Analysis

Currently, DataLex focuses on interactive log analysis, but future improvements can introduce machine learning-based anomaly detection to predict security incidents before they escalate. By utilizing techniques such as unsupervised learning, clustering, and time-series analysis, the system could proactively identify deviations in network behavior that indicate potential threats.

5.2.4 Multi-Language Support and Accessibility

Expansion of natural language processing (NLP) capabilities will enhance the system's accessibility to security professionals across different regions. Future research can focus on multi-language support, allowing security teams to interact with DataLex in their preferred language.

5.2.5 AI-Guided Automated Incident Response

At present, DataLex provides AI-driven security insights, but future iterations can extend its capabilities to automated incident response through Security Orchestration, Automation, and Response (SOAR) functionalities. This would allow DataLex to not only detect security threats but also suggest or execute predefined response actions based on the severity of an attack.

5.2.6 Improving Explainability and Trust in AI-Generated Insights

For security experts to trust AI-driven security assessments, it must be transparent. In order to help analysts comprehend how and why DataLex generates particular security alerts, future improvements should concentrate on explainable AI (XAI) methodologies.

5.2.7 Enhancing the Chat-Based Security Interaction Model

Unlike traditional SIEM solutions, which require manual log analysis, DataLex aims to create an intuitive, chat-based security assistant that allows users to ask complex security questions and receive detailed responses. Future enhancements will make this interaction more advanced by enabling:

- Incident-based troubleshooting

- Alternative solution suggestions
- Problem-Solution Mapping
- Automated security reporting

5.2.8 Cloud and Hybrid Deployment Models

Although DataLex presently uses an on-premises setup to provide the highest level of data protection, hybrid or cloud-based deployment options may be added in later versions. Organizations would benefit from increased scalability and flexibility from this expansion while still adhering to data protection laws. By using cloud-based resources for AI model updates, processing external threat intelligence, and improved computing performance, a cloud-hybrid strategy would enable vital security logs to stay within the organization's local infrastructure.

5.2.9 Evolving into an AI-Powered Security Governance Platform

Beyond its core function of log monitoring and alerting, DataLex has the potential to evolve into a comprehensive cybersecurity governance platform. By integrating compliance enforcement, risk assessment, and security control recommendations, the system could assist organizations in proactively strengthening their security posture. Future enhancements could enable DataLex to analyze historical security incidents and suggest policy improvements tailored to an organization's specific threat landscape.

5.2.10 AI-Driven Security Compliance and Risk Management

For financial institutions and other security-sensitive industries, maintaining regulatory compliance is essential. By continuously monitoring security occurrences,

evaluating risk indicators, and automatically producing audit reports, DataLex can help security teams maintain compliance. The solution could give businesses useful information to match their security policies with recognized standards like GDPR, NIST, and PCI-DSS by incorporating AI-driven analysis.

5.2.11 Conclusion

In order to become a fully interactive, predictive, and self-learning security assistant, DataLex must develop its AI skills beyond basic log analysis. DataLex will offer an end-to-end cybersecurity solution that consistently adjusts to new problems through advancements in automated troubleshooting, proactive defense mechanisms, policy recommendations, and sandboxed threat analysis. With these improvements, DataLex will be positioned well ahead of conventional SIEM solutions and become a cybersecurity ecosystem driven by next-generation AI.

Bibliography

- [1] Victor Jüttner, Martin Grimmer, and Erik Buchmann. ChatIDS: Explainable cybersecurity using generative AI. arXiv preprint arXiv:2306.14504, 2023.
- [2] Ramesh Marri, Sarath Varanasi, and S. V. K. Chaitanya. Integrating security information and event management (SIEM) with data lakes and AI: Enhancing threat detection and response. *Deleted Journal*, 6(1):151–165, 2024. doi: 10.60087/jaigs.v6i1.239.
- [3] Peter Gold, Christopher Moyin, David Samad, Ben Victoria, and Fatimah Rhoda. AI-driven threat intelligence: Enhancing SIEM capabilities for real-time cybersecurity monitoring. 2025.
- [4] K. N. Ovabor, I. O. N. Sule-Odu, T. N. Atkison, A. T. N. Fabusoro, and J. O. N. Benedict. AI-driven threat intelligence for real-time cybersecurity: Frameworks, tools, and future directions. *Open Access Research Journal of Science and Technology*, 12(2):040–048, 2024. doi: 10.53022/oarjst.2024.12.2.0135.
- [5] Mohamed Amine Ferrag, Fatima Alwahedi, Ammar Battah, Bilel Cherif, Abdechakour Mechri, and Norbert Tihanyi. Generative AI and large language models for cybersecurity: All insights you need. arXiv preprint arXiv:2405.12750, 2024.
- [6] C. Strickland, C. Saha, M. Zakar, S. Nejad, N. Tasnim, D. Lizotte, and A. Haque. DRL-GAN: A hybrid approach for binary and multiclass network intrusion detection. arXiv preprint arXiv:2301.03368, 2023.

-
- [7] S. Ethan, R. Samuel, P. Remi, A. Olamide, F. Abdullahi, R. Falade, and F. Adela. Next-gen SIEM and data lakes: Transforming cybersecurity with AI-driven insights, 2025.